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TEAM THE HUSTLERS

Project Title : AI-Powered Mobile Platform for Democratizing Sports Talent Assessment

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Problem Statement :

AI-Powered Mobile Platform for Democratizing Sports Talent Assessment



Aspiring athletes, particularly students and young players from rural areas and economically disadvantaged backgrounds, often lack access to professional coaches and scientific sports assessment facilities. As a result, many talented individuals remain undiscovered due to limited opportunities to demonstrate their abilities.

Existing sports talent assessment methods often rely on manual observation and subjective judgments, leading to inconsistent evaluations. The high cost of professional assessment and the lack of standardized testing make it difficult to compare athletes fairly and identify their strengths and weaknesses.



Therefore, there is a need for an affordable, accessible, and reliable system that enables standardized and transparent sports talent assessment. Addressing this g can help identify promising athletes, provide equal opportunities, and support talen development regardless of geographical location or financial background.

Proposed Solution :

AI-Powered Mobile Platform for Democratizing Sports Talent Assessment

The proposed solution is an AI-powered mobile application designed to identify and assess sports talent among students from diverse backgrounds. It uses smartphone video recordings, computer vision, and AI-based movement analysis to evaluate athletic performance without requiring expensive equipment or specialized facilities. The platform aims to provide accessible, transparent, and performance-based assessment opportunities, particularly for rural and economically disadvantaged students.











Key Features :

- Accessible Mobile Platform: Provides an affordable, user-friendly interface for athletes to perform assessments using smartphones.
- Guided Sports Testing: Offers clear instructions for fitness and sports-specific tests to ensure consistent assessment procedures.
- AI-Powered Movement Analysis: Uses OpenCV and MediaPipe Pose to analyze body posture, joint movements, balance, and coordination.
- Standardized Performance Scoring: Generates scores and grades based on predefined criteria and appropriate age-group benchmarks.
- Automated Performance Reports: Provides reports highlighting athletes' strengths, weaknesses, and areas for improvement.
- Interactive Performance Dashboard: Displays assessment scores, movement analysis, performance trends, and personalized recommendations.
- Progress Tracking: Maintains assessment history to monitor athletic development over time.
- Remote Coach Review: Enables authorized coaches to review athlete reports and identify promising candidates for further evaluation.
- Equal Assessment Opportunities: Reduces geographical and economic barriers, helping rural and urban students showcase their abilities through a more transparent evaluation process.

Methodology / Working Model :



Athlete Registration

Smartphone Video Recording

Video Preprocessing

AI-Based Pose Estimation

Feature Extraction and Performance Scoring

Sports Suitability Assessment

Personalized Feedback and Reports

Database Storage and Progress Tracking

Testing and Validation



Technology Stack :



Backend Tools

- **Python**: core application logic and AI processing
- **SQLite**: stores athlete profiles & assessment records



AI and Computer Vision

- **MediaPipe Pose**: detects body landmarks from video
- **OpenCV**: captures and processes webcam or recorded video
- **NumPy**: calculates joint angles from landmark coordinates



Web Application (Frontend)

Streamlit: interactive interface for assessment and results



Data Management

- **SQLite**: stores athlete profiles & assessment records
- **Pandas**: organizes results and builds the leaderboard

A software-only stack built entirely on free, open-source tools, with no specialized hardware needed.

Innovation and Unique Features :

- AI-Powered Talent Assessment
- Smartphone-Based Analysisgraph text
- Pose Estimation
- Personalized Feedback
- Sports Suitability Suggestion
- Progress Tracking
- Affordable and Accessible



Feasibility, Viability and Impact :

- Current Status: A basic prototype has been developed using Python, Streamlit, OpenCV, MediaPipe Pose, Pandas, and SQLite to demonstrate the project's core functionality
- Technical Feasibility: The project is technically achievable using existing AI tools and open-source technologies.
- Time Feasibility: A minimum of one year is estimated to develop and validate the complete mobile application, including requirement analysis, , data collection, AI model development, expected results will be compared with actual test results to evaluate accuracy, reliability, and fairness and finally deploy the product.
- Economic Feasibility: Open-source technologies help reduce development costs, although data collection, testing, deployment, and maintenance may require additional resources.
- Operational Feasibility: The platform can provide accessible sports assessments, but testing across different devices, environments, and athlete groups is necessary.
- Viability & Impact: The project offers a cost-effective, scalable solution for sports talent identification, with the potential to provide equal assessment opportunities to students regardless of their geographical or economic background.







Expected Outcomes and Conclusion :

- Improved Rural Sports Talent Identification
- Increased Equal Opportunities
- Fair and Transparent Evaluation
- Automated Eligibility Assessment
- AI-Based Generated Performance Reports
- Enhanced Sports Talent Discovery
- Personalized Training Recommendations
- Continious Performance Tracking
- Cost-Effective Solution
- Inclusive Sports Development

The proposed platform aims to make sports talent assessment more affordable, accessible, and consistent by combining smartphone-based testing with AI-assisted analysis.

By reducing barriers to assessment, the system seeks to help aspiring athletes gain better insights into their abilities and improve access to talent development opportunities.

Vision: Every aspiring athlete deserves an opportunity to showcase their potential.

Project GitHub Link :

https://github.com/krishanthini-pixel/sports-talent-assessment

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